

Lecture

Pattern Matching and Machine Learning

for Audio Signal Processing

MA-INF 2212

§0 Organizational Issues

Prof. Dr. Frank Kurth

Permission to record the lecture or parts thereof is *not* granted!

Lecture & Exercise Times and Locations

Lecture: Thursdays, **16:15 s.t. (i.e., sharp!)** – 17:45 h

- The lecture is **regularly held in presence** (HSZ, HS 3)
- In **exceptional cases only** (to be announced separately), the lecture will **be held in Zoom** under

<https://uni-bonn.zoom.us/j/92984663754?pwd=QVIPSk9oNktWT1FNQVJJM0hVR3c2Zz09>

Meeting-ID: 929 8466 3754, Password: 230420

Exercises: Thursdays **18:00** – 19:00 [**First exercises: 30.04.2026**]

- The exercises will be also held in presence (HSZ, HS 3)
- In **exceptional cases only** the exercises will **be held in Zoom** under

<https://uni-bonn.zoom.us/j/94424062258?pwd=SUhHWFB0a0k3MVJ4RkRISmE4S2JUUT09>

Meeting-ID: 944 2406 2258, Password: 159168

Register for the lecture/exam **in BASIS as usual!**

Registration will be open **until May 05th, 2026, 23:55 h**

- Enter the lecture / Zoom **few minutes before the lecture starts**, allowing some time for setting up your own audio equipment (!)
- Everything should work fine as long as **we all keep up some discipline** 😊:
 - Video transmission from participants will be disabled by default
 - Audio transmission from participants will be muted by default
 - **You can unmute your audio manually – please only do so for questions!**
 - Lecture will stop from time to time to allow for questions (QA)
 - Lecture will be (partially) recorded – recording will stop during QA
 - Zoom gives a clear visual indication while the recording is going on, and it automatically plays an audio message when the recording is started or paused.
 - **Please do not transmit audio while the recording is running. In doing so, you automatically consent to having your voice in the recording!**

Exercises & Exercise Sheets

- Participation in the exercises strongly encouraged (1x / week).
- Each week, one exercise sheet has to be solved:
 - exercise sheets distributed by us each week (via WWW),
 - solutions to be handed in via **eCampus**. (Due date is stated on the exercise sheet),
 - will be corrected, scored and returned via email.

Exercise sheets: Only solutions submitted (a) via eCampus und (b) by Friday the latest (see schedule on next page!) are accepted (no exceptions made!)

- You have to work in groups ($2 \leq N \leq 4$ students)
 - Each group hands in a „group solution“ of the exercise sheets.
 - Note however: *Each group member has to be able to demonstrate all of the groups' solutions!*

— Solutions that are obviously copied from other groups will be rejected!

- Exercises coordinated and given by: Mrs. Alessia Cornaggia-Urrigshardt

Schedule: Lectures & Exercise Sheets

	Week			15		16		17		18	
		Thu	Fri	Thu	Fri	Thu	Fri	Thu	Fri	Thu	Fri
	April					16.	17.	23.	24.	30.	
Sheet	out					1		2		3	
	in						-		1		
	scored					-		-		1	
	Week			19		20		21		22	
	May		1.	07.	08.	14.	15.	21.	22.	28.	29.
Sheet	out			4				5			
	in		2		3		4				
	scored			2				3+4			
	Week			23		24		25		26	
	June			04.	05.	11.	12.	18.	19.	25.	26.
Sheet	out					L 08.06.		7		8	
	in				5				6		7
	scored							5		6	
	Week	27		28		29		30			
	July	02. L + E 29.06.	03.	09.	10.	16.	17.	23.	24.		
Sheet	out	9		10*		-		-			
	in		8		9		10*		-		
	scored	7		8		9		10*			

#Exercise sheets: 10

(10*: Preparation for exam)

No lecture/exercises:

14.05., 28.05., 04.06. 

(25.-29.05. excursion week)

**Lectures & Exercises (L+E)
held in Zoom:**

08.06. (no L+E on 11.06.)

Lecture: 16 – 18 h,

No exercises

29.06. (no L+E on 02.07.)

Lecture: 16 – 18 h,

Exercises: 18 – 19 h

Final exam:

18.08.2026, 09:00-11:00 h

Resit exam:

04.09.2026, 10:00-12:00 h

You will be **automatically registered** for the **eCampus** course when registering in BASIS.

In eCampus you will find

- Weekly exercise sheets
- Notes on important dates (exams,...)
- Additional literature as necessary
- Submission site for all exercises (mind the deadlines!)
- Handouts of the slides shown in the lecture (**note however, that additional material is provided during the lecture!**)
- Log in using your **Uni-ID** as user name
- Link to the eCampus course (available also from BASIS):

https://ecampus.uni-bonn.de/ilias.php?baseClass=ilrepositorygui&ref_id=4053118

Obtaining the 6 credit points (CPs)...

Final exam:

Tue., 18.08.26, 9:00-11:00 h [time/location to be confirmed], (90 minutes of writing time)

Exam currently being scheduled in presence (= physically at U Bonn)

Criterion to be admitted to final exam:

Obtaining ≥ 50 % of all possible points/scores in the 10 exercise sheets.

- It is beneficial to actively participate in the exercises!
- If you fail the final exam, a resit exam will be offered on

04.09.2026, 10:00 – 12:00 h [time/location tbc], (90 minutes duration)

- In general: similar amount of points for all sheets
- Total of scores (T) = Sum of possible scores for sheets 1-9
- **Bonus** points for last exercise **sheet 10** (exam preparation)
- Max. available points (M) = Sum of obtained scores for sheets 1-10

- Criterion for exam admission:

$$M / T \geq 0.5$$



Lecturer for Computer Science at CS dept. of U Bonn

Fields of interest:

- Audio Signal Processing
- Multimedia Information Retrieval
- Efficient Algorithms
- Digital Audio and Music Libraries

Member of AG Multimedia Signal Processing (Prof. Clausen)
1997-2006

Current position: Group Head at Fraunhofer FKIE

Contact:

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Staff for this lecture

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- Follow-up of
Foundations of Audio Signal Processing (MA-INF 2113)
(Winter term 2025/26)
- Builds upon the audio part of that lecture.
- More in-depth treatment of signal processing background.
- Applications from
 - Audio Signal Processing
 - Audio and Music Information Retrieval

Seminar Audio MA-INF 2208

- Offered **next winter semester 2026/27**
- Various topics in Audio Signal processing
- Talks will be scheduled mid Jan/early Feb 2027
- **In case of interest, please contact us via email any time during this summer term!**

And now...

... lets go for the contents!